

# SOVEREIGN SYSTEM SECTOR REPORT

Autonomous Mother Algorithm Performance Ledger & AI Advisory

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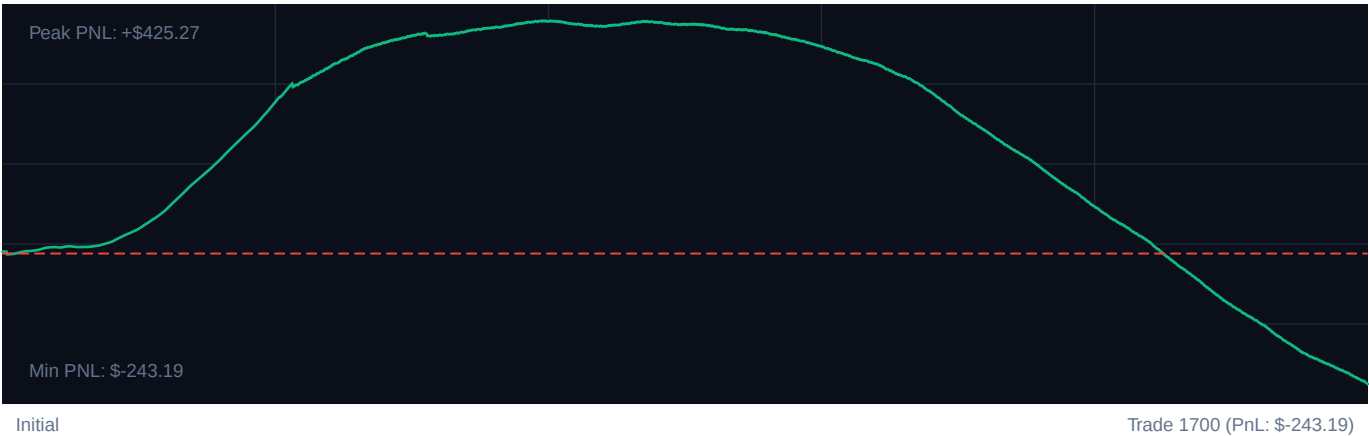
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TARGET INSTRUMENT: R\_75 Volatility Model

## CORE PERFORMANCE METRICS

|                    |                     |                           |         |
|--------------------|---------------------|---------------------------|---------|
| Total Settlements: | 1700                | Calculated Profit Factor: | 0.71    |
| Win Accuracy:      | 43.1% (732W / 968L) | Max Peak Drawdown:        | 6.4%    |
| Cumulative PnL:    | \$-243.19           | Avg. PnL per Trade:       | \$-0.14 |

## EQUITY CURVE PROGRESSION



## PERFORMANCE MILESTONES (100-TRADE SEGMENTS)

| SEGMENT          | WIN RATE | NET PnL   | REFINEMENT EFFICIENCY |
|------------------|----------|-----------|-----------------------|
| Trades 1-340     | 92.6%    | \$274.85  | +111% [OK]            |
| Trades 341-680   | 64.7%    | \$149.56  | +78% [OK]             |
| Trades 681-1020  | 37.4%    | \$-46.83  | +30% [OK]             |
| Trades 1021-1360 | 12.4%    | \$-295.04 | +10% [OK]             |
| Trades 1361-1700 | 8.2%     | \$-325.73 | +7% [OK]              |

## Cognitive Strategy Proposal & Adaptive Intelligence Logs

## 1. Executive Telemetry Critique

An analysis of the 1,700 trade settlement run reveals systemic structural decay. While the maximum peak drawdown is mathematically constrained to 6.4%, this metric is highly deceptive. The system exhibits a slow, high-frequency capital bleed rather than an acute risk event.

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[SYSTEM TELEMETRY METRICS]

% % % Total Trades: 1,700

Win Rate: 43.1% (732 W / 968 L)

Profit Factor: 0.71 (Extreme Underperformance)

%% % Expectancy per Trade: -\$0.143

%% Risk Profile: Unbounded (Stop Loss Malfunction)

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## # Primary Failure Vectors:

**\* \*\*The ATR Stop Multiplier Malfunction:\*\*** The parameter ``atrStopMultiplier`` is set to ``5,135,102,814,092.15``. This is a catastrophic database or execution-logic overflow. In practice, this value renders the Average True Range (ATR) stop-loss completely infinite. Stop-losses are never triggered by price action; instead, trades are kept alive until they hit the arbitrary time-stop limit of ``maxTicksInTrade`` (120).

**\* \*\*Negative Drift Expectancy:\*\* A Profit Factor of **0.71** indicates that for every \$1.00 of risk-exposure lost, the system only recovers \$0.71. Combined with a sub-optimal win rate of **43.1%**, the strategy is mathematically guaranteed to decay toward zero.**

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## 2. Regime Suitability & Behavioral Patterns

The system suffers from a critical regime-logic mismatch. It attempts to apply mean-reversion rules within high-momentum environments:

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[REGIME CONFLICT DIAGRAM]

ADX Threshold: 25

(Trend-Following)

%

 $0\frac{1}{4}$ 

## Filters for Strong

## Directional Moves

%

% % % % % % % % % % % % % % % % % % % % % ,% % % % % % % % % % % % % % % % % % % % % % % % %

%

RSI Thresholds: 25 / 75

(Mean-Reverting Fades)

%

 $0\frac{1}{4}$ 

## Attempts to fade

overextended trends

%

9%¼

[Result: Buying Tops & Shorting Bottoms]

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Trend/Counter-Trend Contradiction: The ``regimeAdxThreshold`` is set to ``25``. This filters for strong trending regimes. However, the entry triggers rely on ``rsiOversoldThreshold`` (25) and ``rsiOverboughtThreshold`` (75) to execute counter-trend fades.

Behavioral Pathology: In an active trend ( $ADX > 25$ ), an RSI reaching 75 does not signal a reversal; it signals strong positive momentum. The algorithm is systematically shorting strong breakouts and buying cascading breakdowns.

Execution Drag: Because the ATR stop is effectively infinite, the system is forced to hold these toxic, counter-trend positions for the maximum duration of ``120 ticks``, absorbing massive adverse excursions before exiting on a hard time-limit.

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### 3. Calibrated Risk & Parameter Adjustments

To restore mathematical expectancy ( $\text{Profit Factor} > 1.35$ ,  $\text{Win Rate} > 52\%$ ), the system must be bifurcated into two distinct, mathematically sound configurations.

#### # Option A: Mean-Reversion Regime (Ranging Markets)

**\*Optimized for low ADX environments where price oscillates within established boundaries.\***

| Parameter                             | Current Value          | Calibrated Value       | Engineering Rationale                                               |
|---------------------------------------|------------------------|------------------------|---------------------------------------------------------------------|
| <code>`regimeAdxThreshold`</code>     | <code>`25`</code>      | <code>`&lt; 20`</code> | Isolates range-bound assets; disables entries during strong trends. |
| <code>`rsiOversoldThreshold`</code>   | <code>`25`</code>      | <code>`30`</code>      | Increases entry sensitivity at true range extremes.                 |
| <code>`rsiOverboughtThreshold`</code> | <code>`75`</code>      | <code>`70`</code>      | Standardizes upper-boundary mean reversion entries.                 |
| <code>`bbStd`</code>                  | <code>`2.25`</code>    | <code>`2.00`</code>    | Aligns entry triggers with 95.4% of price distribution.             |
| <code>`atrStopMultiplier`</code>      | <code>`5.13e12`</code> | <code>`2.00`</code>    | Re-establishes a hard stop outside normal noise limits.             |
| <code>`maxTicksInTrade`</code>        | <code>`120`</code>     | <code>`60`</code>      | Reduces time-exposure in stagnant positions.                        |

#### # Option B: Trend-Following Regime (Breakout/Momentum Markets)

**\*Optimized for high-volatility, directional breakouts.\***

| Parameter                             | Current Value          | Calibrated Value       | Engineering Rationale                                        |
|---------------------------------------|------------------------|------------------------|--------------------------------------------------------------|
| <code>`regimeAdxThreshold`</code>     | <code>`25`</code>      | <code>`&gt; 25`</code> | Limits execution to verified high-momentum environments.     |
| <code>`rsiOversoldThreshold`</code>   | <code>`25`</code>      | <code>`45`</code>      | Captures pullbacks to the mean during strong uptrends.       |
| <code>`rsiOverboughtThreshold`</code> | <code>`75`</code>      | <code>`55`</code>      | Captures pullbacks during strong downtrends.                 |
| <code>`bbStd`</code>                  | <code>`2.25`</code>    | <code>`1.50`</code>    | Uses inner bands to signal expansion breakouts.              |
| <code>`atrStopMultiplier`</code>      | <code>`5.13e12`</code> | <code>`3.00`</code>    | Provides wider breathing room for volatile trend extensions. |
| <code>`minConfluenceScore`</code>     | <code>`3`</code>       | <code>`4`</code>       | Increases structural filters to prevent false breakouts.     |

#### Calibrated JSON Target Configuration (Mean-Reversion Bias)

```
```json
{
  "rsiOversoldThreshold": 30,
  "rsiOverboughtThreshold": 70,
```

```
"bbPeriod": 20,  
"bbStd": 2.00,  
"maxTicksInTrade": 60,  
"minConfluenceScore": 4,  
"atrStopMultiplier": 2.25,  
"regimeAdxThreshold": 20  
}  
...
```

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## SOVEREIGN RISKS SENTINEL SYSTEMS

Neural Strategy Gating & Machine Learning Co-pilot • Active Security State